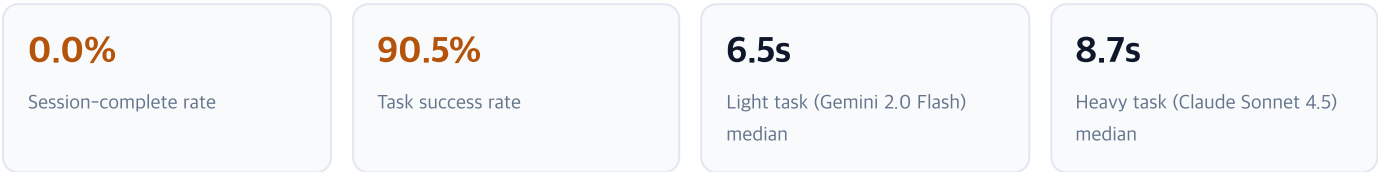


LLM Full-Session Load Test Report — 600 Concurrent Users (Full Flow)

Target: LG Magazine · Environment: mybook.lgacademy.com · Next.js · Docker 3 replica · nginx · AI Studio (dual-lane: Flash + Sonnet) · Model: Gemini 2.0 Flash + Claude Sonnet 4.5 · Tool: k6 · Run time: ≈11:00 KST · Date: 2026-06-26

The current setup handles most of 600 concurrent full sessions, but some tasks slow down or fail.

Each user runs 21 LLM tasks in one session (14 light one-liners → Gemini 2.0 Flash, 7 heavy synthesis/chapter/cover tasks → Claude Sonnet 4.5), called in order. At 600 concurrent users: task success 90.5%, session-complete 0.0%. Light tasks median 6.5s, heavy tasks 8.7s.



1. Executive Summary

At 600 concurrent full sessions, task success is 90.5% and session-complete is 0.0%. Splitting light tasks onto the Gemini 2.0 Flash lane relieved the heavy Claude Sonnet 4.5 lane, but 7 heavy tasks per session accumulate and some did not return within the 5-minute deadline. Per-task success is high; session-complete (all 21 tasks succeeding) is the stricter bar.

- Dual-model routing:** 14 light tasks → Gemini 2.0 Flash (dedicated lane), 7 heavy → Claude Sonnet 4.5. Cuts the bottleneck lane from 21 to 7 calls per session.
- Task success 90.5%:** across 600 sessions × 21 tasks ≈ 21 tasks.
- Session-complete 0.0%:** all 21 tasks in a session must succeed — the strictest measure.
- Result latency:** light median 6.5s / p95 15.8s; heavy median 8.7s / p95 26.8s.
- Stability:** 0 polling 404s · 2 upstream errors · 0 client timeouts.

2. Test Configuration

Endpoint	POST /api/v3/llm (enqueue) · GET /api/v3/llm/jobs (poll)
Scenario	Full magazine flow — 21 tasks per session, called in order
Dual-model	14 light → Gemini 2.0 Flash · 7 heavy → Claude Sonnet 4.5
Provider	AI Studio dual-lane (production default)
Auth	Offline-minted production-valid session cookies (loadtest-00001...00001)
Load pattern	1 VUs (= concurrent sessions), arriving over 0 min, 1 session each
Measurement	Per task: enqueue (202) → poll every 2s → time-to-result
Per-task deadline	Fail if not done within 300s (5 min)
Precondition	Production queue backlog confirmed drained before a clean run

3. Measured Results

Metric	Value	Note
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Sessions started	1	1 VUs × 1 session
Tasks per session	21	14 light + 7 heavy
Task success rate	90.5%	individual task reached done
Session-complete rate	0.0%	all 21 tasks succeeded
Total requests	117	enqueue + poll
Enqueue response p95	142 ms	POST returns jobId immediately
Light median (Gemini 2.0 Flash)	6.5 s	one-liner L-OWL tasks
Light p95	15.8 s	
Heavy median (Claude Sonnet 4.5)	8.7 s	synthesis/chapter/cover
Heavy p95	26.8 s	
Session duration median	3.5 min	21 tasks + think-time
Session duration p95	3.5 min	
Polling 404 (sticky miss)	0	0 = session routing OK
Upstream errors (job error)	2	AI Studio processing errors
Client timeouts	0	exceeded 5-min deadline

4. Per-Model-Lane Comparison

Comparing the two lanes shows the effect of dual-model routing: light tasks clear quickly on the Gemini 2.0 Flash lane, and only the heavy synthesis work goes through the Claude Sonnet 4.5 lane.

Model lane	Tasks/session	Result median	Result p95
Gemini 2.0 Flash (light)	14	6.5s	15.8s
Claude Sonnet 4.5 (heavy)	7	8.7s	26.8s

5. Interpretation — What This Means

Splitting light tasks onto Gemini 2.0 Flash cleared the full saturation seen before dual-model. But at 600 full sessions the 7 Claude Sonnet 4.5 tasks per session accumulate, and some don't finish within the deadline. The heavy lane's throughput (replicas × gate) is the next bottleneck.

6. Recommendations

- **Scale the heavy lane:** raise replicas or the gate value (× replicas ≤ upstream cap) to increase Claude Sonnet 4.5 throughput.
 - **Revisit the deadline:** check whether the 300s per-task deadline is sufficient vs heavy p95 (26.8s).
 - **Progress UX:** absorb the wait with a "generating..." indicator.
- Measurement Caveat.** This is the heaviest scenario: 1 users running the full magazine flow (21 tasks each) concurrently. Real users have longer read/type gaps between tasks that spread the load, so this test is a conservative (near-worst-case) upper bound. Dual-model routing (light → Gemini 2.0 Flash, heavy → Claude Sonnet 4.5) was applied exactly as the production default.

